

Campus Dataset Benchmark Note

Strict Image-Level vs Place-Level Evaluation on the CMU-Africa Campus Dataset

Project Documentation

April 23, 2026

Purpose

This note documents both versions of the campus benchmark:

- the original **strict image-level evaluation** using `test_campus_dataset.py`, and
- the new **place-level multi-positive evaluation** using `test_campus_dataset_place_level.py`.

The goal is to explain why the dataset was relabeled, how the new implementation works, and what changed in the results after moving from strict 1-to-1 matching to place-level matching.

1 Original Dataset and Why It Was Too Strict

The original campus dataset contains:

- **50 day images** as the database,
- **64 night images** as the queries,
- **33 originally matched queries**, and
- **31 original no-match queries** marked with `-npm`, `npmXX`, or `PXL_*`.

In the first version of the evaluation, each night query was expected to match exactly one daytime image. That is easy to implement, but it is too strict for a real campus traversal because multiple day images can show the **same physical place** from slightly different valid viewpoints.

After visually reviewing the query/reference pairs, we found that several so-called false matches were actually:

- the correct place,
- but from a different daytime view,
- and therefore marked as wrong only because the ground truth was 1-to-1.

So the original benchmark mixed together two effects:

- true retrieval errors,
- and correct same-place matches that were unfairly penalized by the labeling.

2 Relabeling Strategy

Core idea

Instead of saying:

one night image $\rightarrow$ one exact day image

the new dataset says:

one night image $\rightarrow$ one **place ID**, which may contain multiple valid day images

What we created

We created a duplicate dataset:

- `custom_dataset_place_level/`

The original `custom_dataset/` was left unchanged for comparison.

The duplicate adds:

- `place_flow.csv`: the ordered sequence of places along the walk,
- `day_place_index.csv`: place ID assigned to each day image,
- `night_rematches.csv`: place-level rematch for each night image.

The relabeling grouped the campus walk into **10 place IDs**, for example:

- P01: atrium central lobby
- P02: red entry transition zone
- P06: red corridor lounge
- P09: outdoor plaza path

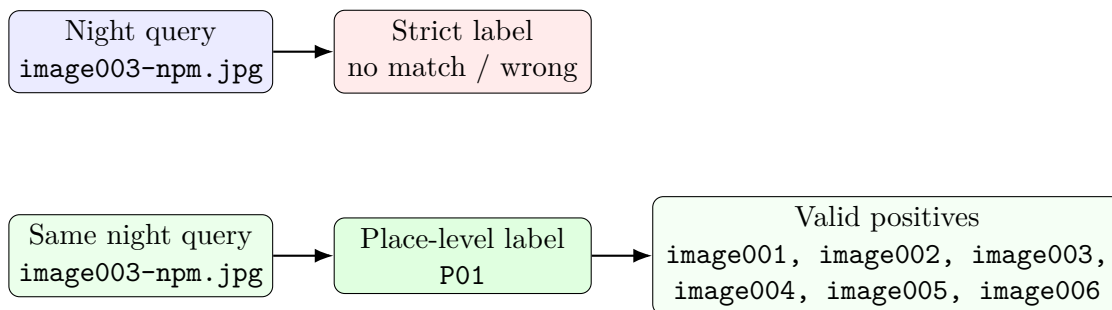


Figure 1: The main relabeling idea: the same query moves from a strict image-level label to a place-level label with multiple valid daytime positives.

Example

One concrete example is the query `image003-npm.jpg`. Under the original labeling it was treated as a no-match image. After visual inspection, it was clearly another night view of the **atrium central lobby**, so in the place-level dataset it is rematched to:

- `place_id = P01`
- valid positives: `image001.jpg | image002.jpg | image003.jpg | image004.jpg | image005.jpg | image006.jpg`

So in the new evaluation, the query is counted as correct if the system retrieves *any* valid daytime image from the same place.

What changed quantitatively

The place-level relabeling produced:

- **64 place-level matched queries**
- **0 remaining no-match queries**
- **12 queries rematched from original -npm labels**
- **10 unique place IDs**

Campus place-level flow: representative day/night views

P01
atrium_central_lobby



day: image001.jpg



night image001.jpg

P02
red_entry_transition_zone

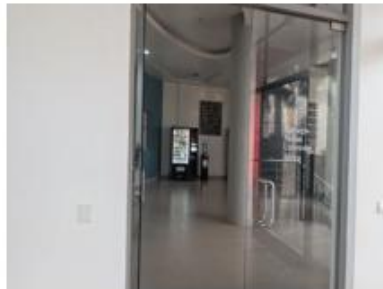


day: image012.jpg

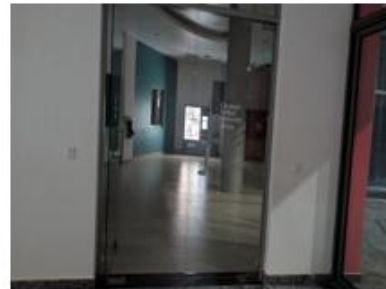


night image012.jpg

P03
innovation_hub_door



day: image019.jpg



night image019.jpg

P04
booth_balcony_walk



day: image021.jpg



night image021.jpg

P05
screen_wall_balcony



day: image025.jpg



night image025.jpg

3 New Implementation

To preserve the original evaluation path, we created new files rather than modifying the old ones:

- original loader: `datasets/load_dataset.py` → `CampusDataset`
- new loader: `datasets/load_dataset_place_level.py` → `CampusPlaceLevelDataset`
- original test script: `test_campus_dataset.py`
- new test script: `test_campus_dataset_place_level.py`

The new loader reads the CSV annotations and builds a **multi-positive** ground truth matrix:

Listing 1: Core idea of the place-level ground-truth construction.

```
valid_names = [
    part.strip()
    for part in row.get("valid_day_images", "").split("|")
    if part.strip()
]
for valid_name in valid_names:
    db_idx = day_lookup.get(valid_name)
    if db_idx is not None:
        gt[db_idx, q_idx] = True
```

So each query column can now contain **multiple true positives** instead of exactly one. Importantly, the new test script still uses the same main evaluation flow:

- descriptor extraction,
- similarity matrix computation,
- thresholded matching,
- precision-recall,
- Recall@K.

That means the comparison between old and new results is still meaningful.

4 Strict Image-Level Results

Descriptor	AUC	R@100P	R@1	R@5	R@10
CosPlace	0.470	0.030	0.727	1.000	1.000
EigenPlaces	0.443	0.000	0.788	1.000	1.000
NetVLAD	0.242	0.000	0.394	0.879	0.939
AlexNet	0.512	0.061	0.758	0.939	0.939
HDC-DELF	0.488	0.030	0.788	0.939	0.970
SAD	0.106	0.000	0.212	0.545	0.727

Table 1: Original strict image-level results from `test_campus_dataset.py`.

5 Place-Level Results

Descriptor	AUC	R@100P	R@1	R@5	R@10
CosPlace	0.627	0.114	0.969	1.000	1.000
EigenPlaces	0.663	0.110	0.922	1.000	1.000
NetVLAD	0.512	0.024	0.891	0.938	0.969
PatchNetVLAD	0.641	0.074	0.938	1.000	1.000
AlexNet	0.387	0.074	0.859	0.969	1.000
HDC-DELF	0.507	0.132	0.922	0.969	0.984
SAD	0.170	0.000	0.297	0.625	0.766

Table 2: Place-level results from `test_campus_dataset_place_level.py`.

6 Direct Comparison

Descriptor	Δ AUC	Δ R@100P	Δ R@1	Main effect
CosPlace	+0.157	+0.084	+0.242	Strong gain once same-place alternate views are credited.
EigenPlaces	+0.220	+0.110	+0.134	Best overall AUC after relabeling.
NetVLAD	+0.270	+0.024	+0.497	Largest recovery in top-1, showing strict labels were penalizing it heavily.
AlexNet	-0.125	+0.013	+0.101	Better top-1, but weaker AUC under place-level evaluation.
HDC-DELF	+0.019	+0.102	+0.134	Much cleaner threshold behavior after relabeling.
SAD	+0.064	+0.000	+0.085	Improves slightly, but remains the weakest model.

Table 3: Change from strict image-level evaluation to place-level evaluation.

7 Main Insights

0. Main conclusion about the original setup

We can now state the conclusion more clearly: the **original campus setup was not robust to multiple valid views under the original evaluation protocol**. In practice, that means:

- the descriptor pipeline was often retrieving the correct *place*,
- but the strict 1-to-1 image labels only credited one exact daytime image,
- so several correct same-place, different-view retrievals were counted as errors.

So the earlier weakness was not only a retrieval problem. It was also an **evaluation-design problem**. The place-level relabeling helps separate:

- truly wrong-place retrievals, from
- correct same-place retrievals that the original setup under-credited.

1. The relabeling changed the story substantially

The place-level results confirm that the original dataset was indeed under-crediting correct retrievals. The clearest examples are:

- **CosPlace**: R@1 rises from **0.727** to **0.969**
- **EigenPlaces**: AUC rises from **0.443** to **0.663**
- **NetVLAD**: R@1 rises from **0.394** to **0.891**

This confirms that many earlier “errors” were not truly wrong places. They were often **same-place, different-view retrievals**.

2. EigenPlaces and CosPlace remain the strongest overall retrievers

Under the place-level protocol:

- **CosPlace** gives the best **R@1 = 0.969**
- **EigenPlaces** gives the best **AUC = 0.663**
- both keep **R@5 = 1.000** and **R@10 = 1.000**

So the broad conclusion from the earlier benchmark remains true, but it is now supported by a fairer labeling protocol.

3. PatchNetVLAD becomes much more credible at place level

PatchNetVLAD was not part of the original comparison table, but in the place-level evaluation it performs strongly:

- **AUC = 0.641**
- **R@1 = 0.938**
- **R@5 = 1.000**

This suggests PatchNetVLAD benefits from being evaluated on actual place consistency rather than exact image duplication.

4. AlexNet does not benefit as much as the others

AlexNet improves in **R@1**, but its place-level **AUC** drops from **0.512** to **0.387**. So while it still gives a decent top-1 rate, it is no longer the most appealing option once the relabeling issue is corrected.

5. HDC-DELF looks more stable after relabeling

Previously HDC-DELF looked noisy because many thresholded matches were marked as false positives. Under the place-level protocol:

- **R@100P** rises from **0.030** to **0.132**
- the printed threshold statistics no longer show false positives in the current run

So HDC-DELF appears more reasonable once same-place alternative views are credited correctly.

6. SAD remains weak

SAD improves slightly, but it is still clearly the weakest descriptor in both protocols.

8 How To Interpret the New Metrics

The place-level evaluation is better aligned with real localization, but it is also important to recognize what changed:

- there are now **0 remaining unmatched queries**,
- so the new benchmark is no longer testing open-set rejection in the same way,
- instead, it is mainly testing **place-level retrieval quality**.

So the two protocols serve different purposes:

- **strict image-level benchmark**: harsher, but still useful for exact pair evaluation
- **place-level benchmark**: better for real localization quality

9 What This Means for the Project

The main conclusion is now clearer:

- the campus benchmark was not just exposing descriptor weakness,
- it was also exposing a **ground-truth mismatch** between image-level labels and place-level localization.

After correcting that mismatch, the strongest methods are:

- **CosPlace**: strongest top-1 retrieval
- **EigenPlaces**: strongest overall AUC
- **PatchNetVLAD**: strong additional option for place-level retrieval

10 Recommended Next Steps

1. Keep **both** evaluation paths in the repo:
 - strict image-level for comparison with the original protocol
 - place-level for realistic localization analysis
2. Use the **place-level benchmark** when arguing about real retrieval quality.
3. Use the **strict benchmark** carefully, mainly to show how much the labeling assumption matters.
4. If open-set rejection still matters, add a third benchmark split with truly unmatched night queries that remain unmatched even after place-level review.
5. For the live demo, prioritize **CosPlace**, **EigenPlaces**, or **PatchNetVLAD**.

Bottom Line

The place-level relabeling was worth doing. It shows that the original campus setup was **not robust to multiple valid views under a strict image-level evaluation**, because it forced a 1-to-1 image correspondence where the real task is **place recognition across viewpoint changes**. After the relabeling, the strongest descriptors improve substantially, especially CosPlace, EigenPlaces, NetVLAD, and HDC-DELF. The project can now report a much fairer campus evaluation while still keeping the original strict benchmark for comparison.